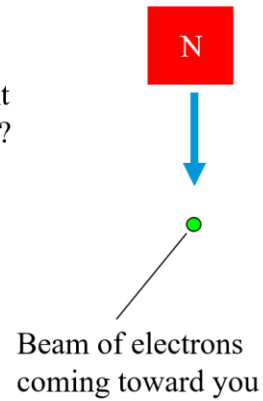


Q27.1

A beam of electrons (which have negative charge q) is coming straight toward you. You put the north pole of a magnet directly above the beam. The magnetic field from the magnet points straight down. Which way will the electron beam deflect?

- A. upward
- B. downward
- C. to the left
- D. to the right
- E. It won't deflect at all.



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Q27.5

A positively charged particle moves in the positive z -direction. The magnetic force on the particle is in the positive y -direction. What can you conclude about the x -component of the magnetic field at the particle's position?

- A. $B_x > 0$
- B. $B_x = 0$
- C. $B_x < 0$
- D. not enough information given to decide

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Q27.10



A charged particle moves through a region of space that has both a uniform electric field and a uniform magnetic field. In order for the particle to move through this region at a constant velocity,

- A. the electric and magnetic fields must point in the same direction.
- B. the electric and magnetic fields must point in opposite directions.
- C. the electric and magnetic fields must point in perpendicular directions.
- D. The answer depends on the sign of the particle's electric charge.

Nombres: